实数

Notation: $\mathbb{R}$: the set of real numbers

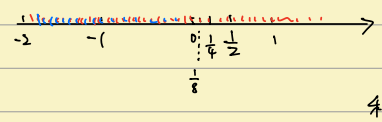
$\mathbb{Q}$: the set of rational numbers 有理数

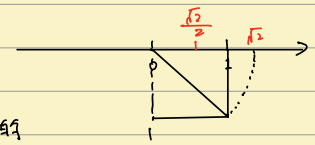
$x \in \mathbb{Q}$: $x = \frac{p}{q}$ $(q \neq 0)$ p, q integer numbers 整数

$\mathbb{R}$: $\mathbb{Q} \cup \mathbb{I}$ $\qquad$ $\mathbb{I}$: irrational number

$\sqrt{2}$ $\quad 1+\sqrt{2}$ $\quad \frac{\sqrt{2}}{3}$

$0.\dot{9} = 1$ $\qquad 0.999\ldots = 1$





稠密的

The set of rational numbers is dense

The set of real number is continuous 连续的

1.1 Some set theory

Definition

Defn

Defn: A set is a collection of different things.

Properties: ① Well-definedness 确定性 : $a \in A$ or $a \notin A$

② Distinct elements 互异性 :

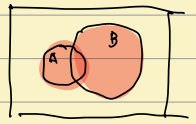
③ Unordered 无序性

Notation: $a \in A$ $\quad a \notin A$

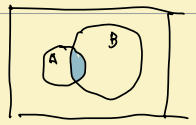
$A \subset B \Leftrightarrow \{A \subseteq B$ and $A \neq B\}$ $\quad B \supset A$ $\quad B \supseteq A$

Property: if $A \subseteq B$ and $B \subseteq A$ if and only if $A = B$

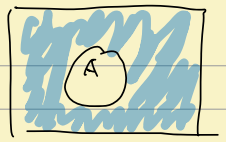
Union 交集: $A \cup B = \{x \in \mathbb{R} : x \in A$ or $x \in B\}$



intersection 并集: $A \cap B = \{x \in \mathbb{R} : x \in A$ and $x \in B\}$



complement 补集: $A^c = \{x \in \mathbb{R} \mid x \notin A\}$



Empty set: 空集 a set that contains no element.

指标集

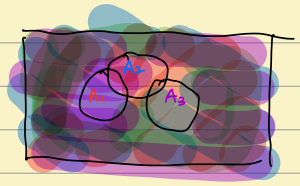
$A_1, A_2, A_3 \cdots A_n$ $\quad I = \{1, 2, 3, \dots, n\}$

$$A_1 \cup A_2 \cup A_3 \cdots \cup A_n = \bigcup_{i \in I} A_i := \{x \in \mathbb{R} \mid x \in A_i \text{ for some } i\}$$

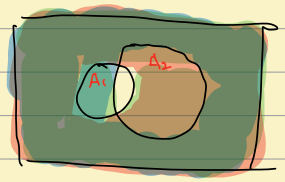
$$\bigcap_{i \in I} A_i := \{x \in \mathbb{R} \mid x \in A_i \text{ for all } i\}.$$

De Morgan's Laws

① $$\left(\bigcup_{i \in A} A_i\right)^c = \bigcap_{i \in I} (A_i^c)$$



② $$\left(\bigcap_{i \in A} A_i\right)^c = \bigcup_{i \in I} (A_i^c)$$



1.2 Intervals 区间

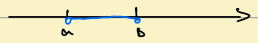
An interval is a subset of the set of real numbers $\mathbb{R}$:
given arbitrarily real numbers a and b such that $a < b$

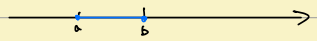
finite interval

$]a, b[$

Notations: $(a, b) := \{x \in \mathbb{R} \mid a < x < b\}$ open interval

$[a, b] := \{x \in \mathbb{R} \mid a \le x \le b\}$ closed interval



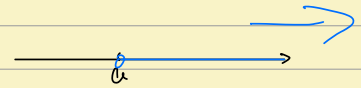


$[a, b) := \{x \in \mathbb{R} \mid a \le x < b\}$

$(a, b] := \{x \in \mathbb{R} \mid a < x \le b\}$

infinite intervals

$(a, \infty) := \{ x \in \mathbb{R} \mid x > a \}$



$[a, \infty) := \{ x \in \mathbb{R} \mid x \geq a \}$

$(-\infty, b) := \{ x \in \mathbb{R} \mid x < b \}$

$(-\infty, b] := \{ x \in \mathbb{R} \mid x \leq b \}$

$(-\infty, +\infty) := \mathbb{R}$

$\mathbb{R}_+ := [0, +\infty)$ $\quad$ $\mathbb{R}_{++} := (0, +\infty)$

Example 1: Given $x, y \in \mathbb{R}$, $x < y$, then there exists a rational number r, such that

$$x < r < y$$

Proof:

Since $x < y$, then there exists $n > 0$ such that $\bar{x}_n < y_n$, let $r = \frac{1}{2}(\bar{x}_n + y_n)$

$\Rightarrow$ $x \leq \bar{x}_n < r < y_n < y$

We conclude $x < r < y$ and r is a rational number.

x: $a_0.a_1a_2a_3 \cdots a_n \cdots$

e.g. $x = 11.347165$

x_n with $n = 4$

x 是近似 $x_n = 11.3471$

过剩近似 $\bar{x}_n = 11.3472$ $\quad$ $\bar{x}_n > x_n$

Example 2: Given $x, y \in \mathbb{R}$, if for any $\varepsilon > 0$, we $x < y + \varepsilon$, then $x \leq y$.

Proof: By contradiction. If $x > y$, let $\varepsilon = x - y$. Since $\varepsilon > 0$, and $x = y + \varepsilon$ Contradiction.

Properties of the set of the real numbers.

① Closure property: 四则运算封闭性 Any arithmetic operation $(+, -, \times, \div)$ is performed on any two numbers of the set, with the answer being another number from the set itself.

② Ordered: 有序性 Any two elements from $\mathbb{R}$, one of the following holds:

$a < b$, $a > b$, $a = b$

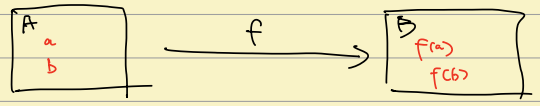
③ Transitivity 传递性: if $a>b$, $b>c$, then $a>c$

④ Density 稠密性: if $a<b$, then there exists c such that $a<c<b$

1.3 Functions 函数

single-value function

Multi-value function 多值函数



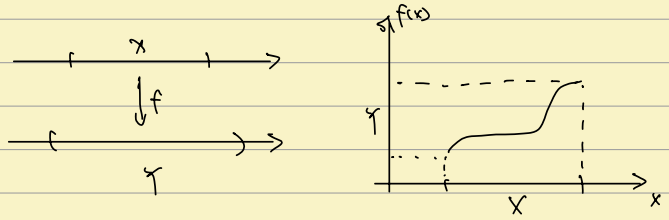
Real function: $f: X \to Y$ with $X, Y \subseteq \mathbb{R}$

X: the **domain** of f. $Dom(f)$ 定义域

Y: the **range** of f. $Im(f)$ 值域 codomain, 像 image

domain 领域

range 范围

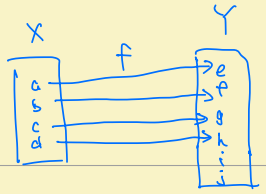


The graph 图 of f: $\{(x, f(x)) \in \mathbb{R}^2 : x \in X\}$

Example 1: $f(x) = 1$, $x \in \mathbb{R}$ $\quad$ $g(x) = 1$, $x \in \mathbb{R} \setminus \{0\}$

$Dom(f) = \mathbb{R}$ $\quad Im(f) = \{1\}$ $\quad$ $Dom(f) = \mathbb{R} \setminus \{0\}$ $\quad Im(f) = \{1\}$

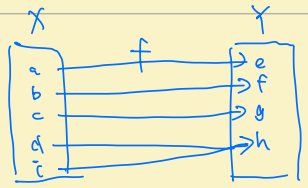
Example 2: $\varphi(x) = |x|$, $x \in \mathbb{R}$ $\quad$ $\psi(x) = \sqrt{x^2}$ $\quad x \in \mathbb{R}$



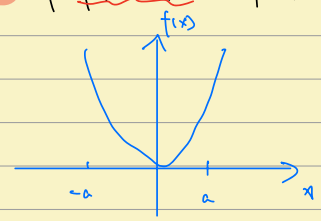
Defn 1.2: Let $f: X \to \mathbb{R}$ be a real function.

The function f is said to be **injective** 单射 (or one-to-one) if for any a and b in X, $f(a) = f(b)$ implies $a = b$.

Equivalently if $a \neq b$ then $f(a) \neq f(b)$. The function f is called **Injection** 单射函数



2. The function f is said to be surjective (满射) if $f(X) = Y$. The function f is called a surjection.

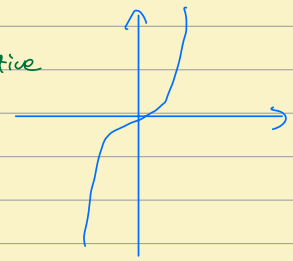


Example 3: $f: \mathbb{R} \to [0, \infty)$, by $f(x) = x^2$

Not an injection.

But it is a surjection.

3. The function f is said to be bijective (双射) if it is both injective and surjective. Then function f is called a bijection.



Example 4. $f: \mathbb{R} \to \mathbb{R}$, by $f(x) = x^3$ bijective

Example 5. $f: [0, \infty) \to \mathbb{R}$, by $f(x) = \sqrt{x}$

injective

Operations of functions.

Given real functions $f_1: X_1 \to Y_1$ $f_2: X_2 \to Y_2$ $X = X_1 \cap X_2$

if $X \neq \phi$

$$F(x) = f_1(x) + f_2(x), \qquad F(x): X \to Y$$

$$G(x) = f_1(x) - f_2(x),$$

$$H(x) = f_1(x) \cdot f_2(x)$$

Let $X^* = X_1 \cap \{x \mid f_2(x) \neq 0, \text{ and } x \in X_2\} \neq \phi$

$$L(x) = \frac{f_1(x)}{f_2(x)} \qquad L(x): X^* \to Y$$

Example 5: $X \neq \phi$ in necessary. $f_1(x) = \sqrt{1 - x^2}$ $f_2(x) = \sqrt{x^2 - 4}$

$F(x) = f_1(x) + f_2(x) = \sqrt{1 - x^2} + \sqrt{x^2 - 4}$ is not well-defined

composed 合成

复合函数

Composed function

Given $y = f(u)$, $u \in D$, $u = g(x)$, $x \in E$

Let $E^* = E \cap \{g(x) \in D\}$. If $E^* \neq 0$ then the composed function is denoted as

$$y = f(g(x)) \qquad x \in E^*$$

Example 6. $y = f(u) = \sqrt{u}$, $u = g(x) = 1 - x^2$

$D = [0, +\infty)$ $\qquad E = \mathbb{R}$

$y = f(g(x)) = \sqrt{1 - x^2}$

$$\begin{aligned} E^* &= E \cap \{g(x) \in D\} \\ &= \mathbb{R} \cap \{g(x) \geq 0\} \\ &= [-1, 1] \end{aligned}$$

反函数

Inversed function

Given a real function $f: X \to Y$ by $y = f(x)$, the inversed function of f is

$$f^{-1}: f(X) \to X \text{ by } x = f^{-1}(y) \qquad y \in f(X)$$

Example $\underline{y = x^2}$ $\Rightarrow$ $x = \sqrt{y}$ $\qquad x = -\sqrt{y}$

$X = \mathbb{R}$ $\quad Y = [0, +\infty)$ $\quad [0, +\infty)$.